

ALEX CLONAN

Storrs, CT | alex.clonan@uconn.edu | [LinkedIn](#) | aclonan.github.io

QUALIFICATIONS

- Six years of experience in signal processing methods implemented in computational auditory neuroscience research
- Strong background in data analysis, stimulus design methods for psychophysics and analytical algorithm development
- Proficiency in MATLAB, Python, various machine learning toolboxes, and experience with real-time audio hardware

EDUCATION

UNIVERSITY OF CONNECTICUT

Bachelor of Science in Engineering and Bachelor of Science (Dual Degree)

Storrs, CT

Dec 2022

Major in Electrical Engineering *Dean's List*

Member of Eta Kappa Nu, the International Honor Society for Electrical and Computer Engineers

Major in Molecular Cell Biology *Dean's List, Honors Program*

Thesis: Developing a Physiologically Inspired Model of Human Speech Recognition in Noise

Doctor of Philosophy, Biomedical Engineering, *Concentration Neural Engineering*

in progress

Advisor: Dr. Monty Escabi & Dr. Ian Stevenson

Aug 2023 – Present

Relevant Coursework: Psychoacoustics, Auditory Brainstem Responses, DSP, Machine Learning, Optimization Theory

RESEARCH EXPERIENCE

Escabi Physiological Acoustics Lab (Undergrad 2020-2023, Graduate 2023-Present)

Storrs, CT

Research Assistant

Oct 2020 – Present

- Administered psychoacoustic experiments on speech recognition in natural environmental noise
- Collected in vivo neural recordings from the rabbit inferior colliculus, evoked response to auditory stimuli
- Applied generalized perceptual regression, linking salient integrated acoustic cues to behavioral performance
- Developed a generative sound synthesis model using frequency, temporal modulation and spectral modulation bands
- Completed open-data analysis exploring the acoustic cues underlying sound aversion perception for misophonia

Center for Quantitative Medicine

Farmington, CT

Undergrad Research Fellow Health Research Program

May 2019 – Oct 2020

- Implemented a python-based, neural network simulation of cell signaling motifs in the immune system
- Generalized programming routines for use in different immune response signal transduction pathways

RESEARCH INTERESTS

- Human auditory electrophysiology, related to theories of natural sound perception (Speech, Music, Reverb, etc.)
- Novel stimulus design methods, using generative models of sound for targeted hypothesis testing
- Statistical generative models (spectrum, modulation) and fine structure/envelope recovery (Griffin-Lim) perception
- Interpretable methods to contrast machine and human auditory perception of acoustic stimuli
- Speech enhancement in real-time audio hardware, through perceptually optimized spectrotemporal filter design

TEACHING EXPERIENCE

UConn Department of Biomedical Engineering

Storrs, CT

Teaching Assistant

August 2023 – Present

- Instructed over 100 students each semester about signal processing methods, Fourier analysis and filter design
- Led a lab section teaching students about biomedical signal sensors, data collection and analysis methods
- BME 3400 (Biomedical Signal Processing), BME 3500 (Biomedical Engineering Measurements)

GRADUATE LEADERSHIP EXPERIENCE

Office of Engineering Experiential Education
Graduate Mentor, Instructor and Project Lead

Storrs, CT
August 2022 – Present

- Led and supervised 25 student leaders over 4 years, developing events for community and academic engagement
- Gave presentations promoting narratives of DEI and community engagement for first-year engineering students
- Implemented, and secured via grant a curriculum targeted towards social impact and awareness of engineering education
- Taught and graded sections of UNIV 3820, the second-year engineering seminar, focused on experiential education

PUBLICATIONS

Clonan AC, Zhai X, Stevenson I, Escabí MA Interference of mid-level sound statistics predicts human speech recognition in natural environmental noise *Journal of Neuroscience* 2025

Clonan AC, Stevenson I, Escabí MA I Identifying the acoustic fingerprints of trigger sounds and predicting discomfort for misophonia *Hearing Research* 2026

Clonan AC, Stevenson I, Escabí MA Modulation statistics allow robust prediction of speech recognition accuracy across many words, voices, and natural background sounds (In Review)

Clonan AC, Stevenson I, Escabí MA The perceptual invariance of the temporal fine structure (In Prep)

AWARDS

Health Research Program (HRP) Award, 2019
UConn Co-op Legacy Fellowship, 2020 - *Pediatric Setting Engineering Education*
UConn Summer Undergraduate Fund Research Award, 2022
UConn Conference Presentation Award, Fall 2022
BME Travel Award, Spring 2023, Fall 2023, Spring 2024, Fall 2024
Next Gen Scholar Graduate Fellowship, Fall 2023
IBACS Outreach Fellowship, Spring 2024
NSF GRFP Honorable Mention, Spring 2024

SKILLS

Programming Languages: MATLAB, Python, R, Scheme
Tools & Libraries: PyTorch, TensorFlow, Keras, scikit-Learn
Signal Processing: Filter Design, Generative Audio Models
Optimization Theory: Bayesian, Convex, Numerical
Electrophysiology: In Vivo Neural Recordings from IC
Psychophysics: Paradigm Building and Data Collection

CORE COMPETENCIES

Signal Processing for Audio and Speech, Statistical Analysis, Feature Extraction, Algorithm Development and Acoustics

PROFESSIONAL MEMBERSHIPS

Society for Neuroscience
Association for Research in Otolaryngology

SELECTED RECENT & RELEVANT PRESENTATIONS (4 of 15)

Clonan AC, Invited Talk, From Neural Computations to Practical Intervention: Brain-Inspired Diagnostics and Personalized Sound Correction for Real-World Hearing, March 26 2026

Clonan AC, Zhai X, Stevenson I, Escabí MA, Building Better Hearing Prostheses: Brain-Inspired Diagnostics and Personalized Sound Correction for Real-World Hearing, Poster Presentation at the UConn Technology Showcase, September 2025

Clonan AC, Zhai X, Stevenson I, Escabí MA, Using Word-Level Acoustic Interference of Spectrum and Modulation Statistics to Predict Speech Recognition Behavior in Natural Environmental Noise, Poster Presentation at the Society for Neuroscience/Advances and Perspectives in Auditory Neuroscience, Chicago, IL, October 2024

Clonan AC, Zhai X, Stevenson I, Escabí MA, Interference of mid-level sound statistics predicts speech recognition in natural environmental noise, Oral Presentation at the Association for Research in Otolaryngology, Anaheim, CA, February 2024